SimGate: Safety-Aware Routing to Deterministic Computation for LLM Agents in Rule-Governed Environments

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Abstract

LLM agents operating in rule-governed environments suffer catastrophic failures from hallucinated state transitions—e.g., predicting boiling at incorrect temperatures or forgetting prior state changes. We propose **SimGate**, a modular framework that learns *when* to route agent actions to deterministic external computation (environment simulators) via a lightweight explicit router. A 1.7B-parameter router, trained with supervised fine-tuning on counterfactual trajectory labels, selectively invokes the simulator at 34.2% of steps. On ScienceWorld (18 task types), SimGate achieves a score of 90.2 under the CogRouter protocol (vs. CogRouter 84.6, ITP-R 62.58), reducing catastrophic failures by 72.3% compared to pure LLM reasoning while using 58% fewer tokens than unconditional simulation. An explicit-vs-implicit ablation shows the modular router outperforms backbone-entangled routing by 11.4 points. Analysis of per-task routing patterns reveals a clear failure taxonomy: temperature physics hallucinations and state persistence errors account for 56% of all routed steps, demonstrating the router's utility as a diagnostic probe of systematic LLM weaknesses.

1 Introduction

LLM-based agents have shown impressive capabilities in interactive environments Yao et al. (2023); Shinn et al. (2023), yet they remain prone to *catastrophic failures* in domains governed by deterministic rules. In ScienceWorld Wang et al. (2022), an agent that hallucinates a temperature transition (e.g., predicts water boils at 80°C) triggers irreversible death penalties of -100, collapsing episode scores regardless of prior progress. Prior work addresses this via approximate world models Hao et al. (2023) or implicit tool-use signals embedded in the backbone Gou et al. (2024), but no method provides an *explicit, modular* router that learns *when* deterministic computation is needed.

ITP-R Qiu et al. (2024) routes to an approximate LLM world model but underperforms pure internal reasoning on ScienceWorld (62.58 vs. CogRouter 84.6). CogRouter Qian et al. (2024) uses cooperative RL to select among reasoning modes but does not distinguish between LLM-internal and deterministic-external computation. ReTool Feng et al. (2025) and AgentMath Zhang et al. (2024) learn code-interpreter invocation but entangle routing decisions with action generation in the same forward pass.

We propose **SimGate**, a three-component framework consisting of (1) an LLM agent backbone, (2) a deterministic environment simulator serving as a "safety net," and (3) a lightweight explicit router trained to detect high-risk states. Our contributions:

- **C1**: We show that learned selective routing to deterministic computation achieves 90.2 on ScienceWorld (CogRouter protocol), with the primary value mechanism being catastrophic failure prevention—death avoidance contributes approximately 70% of the gain over pure LLM reasoning.
- **C2**: A 1.7B explicit router, architecturally decoupled from the 7B backbone, outperforms implicit backbone-entangled routing by 11.4 points, and its per-task routing patterns yield an interpretable taxonomy of systematic LLM failure modes.

2 Method

SimGate operates in three stages: counterfactual data collection, router training, and inference-time routing.

Stage 1: Counterfactual Trajectory Collection. We run a Qwen2.5-7B agent on ScienceWorld across 18 task types (20 episodes each, 360 total, max 30 steps per episode). At each decision step, we *fork* execution into three modalities: (a) **fast-internal** (direct LLM response), (b) **deep-internal** (chain-of-thought reasoning), and (c) **deterministic** (environment simulator). Each fork runs to completion, and *hindsight labeling* assigns the best-performing modality as the ground-truth label for that (s_t, o_t) pair. This yields $\sim$8,200 labeled tuples across all task types.

Label distribution: fast-internal 38%, deep-internal 24%, deterministic 38%. Among deterministic labels, 72% correspond to *death avoidance* (the agent's original action would incur -100 penalty) and 28% to *positive guidance* (the simulator corrects erroneous state reasoning without preventing death).

Stage 2: Router Training. We fine-tune Qwen3-1.7B as a 3-class router via SFT on the counterfactual labels. Input: concatenation of interaction history and current observation. Output: one of $\{$`fast-internal`, `deep-internal`, `deterministic`$\}$. Training uses an 80/20 episode-level split ($\sim$6,560 train / $\sim$1,640 val), learning rate 2×10^{-5}, batch size 32, 3 epochs, max sequence length 2048. The router converges at epoch 2.4 with validation accuracy 81.3% (deterministic class: 84.7%; internal classes: 78.9%; macro F1: 0.79).

Stage 3: Inference-Time Routing. At each step, the router classifies the current state and routes to the predicted modality. The simulator is invoked only when the router predicts `deterministic`, yielding a 34.2% invocation ratio. This selective strategy avoids the context bloat and token overhead of unconditional simulation.

3 Experiments

3.1 Setup

We evaluate on ScienceWorld Wang et al. (2022) using the CogRouter evaluation protocol Qian et al. (2024) across 18 task types. The agent backbone is Qwen2.5-7B; the router is Qwen3-1.7B. We report bootstrap 90% confidence intervals where applicable.

3.2 Main Results

Table 1 presents end-to-end scores. SimGate achieves 90.2 [87.1, 93.3], surpassing CogRouter (84.6) and ITP-R (62.58). The gain over always-internal (52.8) is 37.4 points [31.2, 43.6], of which approximately 26 points stem from death avoidance: always-internal suffers a 42.3% catastrophic failure rate, which SimGate reduces to 11.7% (72.3% reduction). The remaining $\sim$11 points come from positive guidance correcting state reasoning in 4–5 task types ($\sim$6 points) and avoiding non-fatal but inefficient hallucinated actions ($\sim$5 points).

Always-deterministic (73.4) underperforms SimGate by 16.8 points despite having access to the simulator at every step. Unconditional simulation introduces context bloat and disrupts the agent's reasoning on steps where LLM-internal computation suffices (e.g., navigation, object manipulation). SimGate uses 58% fewer LLM tokens than always-deterministic.

The oracle router achieves 93.1, leaving only a 2.9-point gap to SimGate—the learned router approaches optimal routing. Random routing (64.5) and a confidence-threshold heuristic (71.8) perform substantially worse, confirming that effective routing requires learned state-dependent decisions.

Method	Score	Cat. Fail %
Always-internal	52.8	42.3
Random routing	64.5	–
Confidence heuristic	71.8	–
Always-deterministic	73.4	–
Implicit routing	78.8	24.1
ITP-R†	62.58	–
CogRouter†	84.6	–
SimGate	**90.2**	**11.7**
Oracle routing	93.1	–

Table 1: End-to-end ScienceWorld scores (CogRouter protocol). Cat. Fail = catastrophic failure rate (%). † = scores from published papers.

	Score	Sim %	FN %
Implicit	78.8	52.3	18.7
Explicit (SimGate)	**90.2**	**34.2**	**8.3**
Δ [90% CI]	+11.4 [7.2, 15.6]		

Table 2: Explicit vs. implicit routing. Sim% = simulator invocation ratio; FN% = false negative rate (missed high-risk steps).

Per-Task Breakdown. SimGate's gains are concentrated in physics and chemistry tasks. On `boil`, the gain is largest: SimGate 95.2 vs. internal 38.4 (+56.8). Similar patterns hold for `melt` (93.8 vs. 41.2), `change_state` (94.1 vs. 39.5), and `freeze` (92.7 vs. 43.8). On simple navigation/manipulation tasks (`pick_up`, `move_to`, `open_door`), internal baselines already exceed 80 and SimGate adds < 3 points; 14 of 18 task types show significant gains.

3.3 Explicit vs. Implicit Routing

We compare SimGate's explicit modular router against an implicit baseline where the 7B backbone is SFT-trained to emit a `[SIMULATE]` token at oracle-labeled steps, entangling routing and action generation in a single forward pass. Both use identical training data.

Table 2 shows the explicit router outperforms implicit routing by 11.4 points [7.2, 15.6]. Three factors drive this gap: (1) Implicit routing entangles routing and action generation in the same forward pass, degrading action quality (catastrophic failure: 24.1% vs. SimGate 11.7%). (2) The backbone over-invokes the simulator (52.3% vs. 34.2%), having learned a "simulate is safer" bias without precise risk assessment. (3) Critically, implicit routing has a higher false negative rate (18.7% vs. 8.3%)—it misses genuinely high-risk steps because backbone attention is diluted by action generation.

3.4 Routing Pattern Analysis

We analyze the router's decisions across 18 task types and 3 episode phases (early / mid / late) to characterize *when* and *why* the router invokes deterministic computation.

Task Clusters. Routing patterns form four clusters: (1) **Physical state change** (boil, melt, freeze, change-state): 55–65% deterministic routing, concentrated in mid-phase. (2) **Chemical reaction** (mix-paint, react-chemicals): 40–50%, mid-to-late phase. (3) **Biological growth** (grow-plant, use-fertilizer): 30–40%, spread across phases. (4) **Navigation/manipulation** (move-to, pick-up, open-door): 5–12%, no phase concentration.

Router	Acc.%	Score	Lat. (ms)
Qwen2.5-1.5B	76.8	86.9	12
Qwen3-1.7B	**81.3**	**90.2**	14
Qwen2.5-3B	82.1	90.8	23

Table 3: Router size ablation. Lat. = per-step inference latency.

Phase Distribution. Deterministic routing peaks in mid-phase (48.7%), where physical and chemical operations occur. Early-phase routing is low (18.4%; mostly navigation/setup), and late-phase intermediate (32.9%; state already determined for many tasks).

Failure Taxonomy. We categorize the steps where the router invokes deterministic computation into five failure modes: **FM1**—temperature physics hallucination (34%; e.g., predicting boiling at wrong temperature); **FM2**—state persistence error (22%; forgetting prior state changes); **FM3**—chemical combination error (18%; wrong reaction products); **FM4**—multi-step causality skip (15%; omitting prerequisite steps); **FM5**—quantity tracking loss (11%; losing track of volumes). FM1 and FM2 together account for 56% of all routed steps, both corresponding to well-documented LLM weaknesses in numerical reasoning and long-horizon state tracking.

Positive Guidance. Beyond death avoidance, the simulator provides positive guidance—correcting erroneous intermediate reasoning even when the error would not cause death—on 4 task types (`grow-plant`, `use-thermometer`, `mix-paint`, `react-chemicals`), contributing 28% of deterministic labels.

3.5 Additional Ablations

2-Way vs. 3-Way Routing. A binary router (internal vs. deterministic) scores 87.6, 2.6 points below 3-way SimGate (90.2; p=0.08, marginal). The binary router defaults to deep reasoning for all internal steps, consuming 18% more tokens. The core value lies in the simulate-or-not decision; 3-way routing adds compute efficiency without significant score improvement.

Router Size. We compare Qwen2.5-1.5B, Qwen3-1.7B, and Qwen2.5-3B as routers (Table 3). The 1.7B model is the performance–efficiency sweet spot: the 1.5B model loses 3.3 SimGate points (primarily from increased false negatives on subtle state conflicts), while the 3B model gains only 0.6 points at 64% higher inference latency.

Cross-Task Transfer. Training the router on chemistry tasks only and evaluating on physics tasks yields 73.2% accuracy (vs. 81.3% full); the reverse gives 70.8%. The 8–10% gap reflects domain-specific failure modes, but 62% of failure modes are shared across domains (FM1, FM2), enabling moderate cross-task transfer. SimGate scores under transfer are 85.4 (chem→phys) and 83.1 (phys→chem), both still exceeding CogRouter.

4 Conclusion

SimGate demonstrates that safety-aware routing to deterministic computation is a highly effective strategy for LLM agents in rule-governed environments. A lightweight 1.7B router, trained on counterfactual trajectory labels, reduces catastrophic failures by 72.3% and achieves 90.2 on ScienceWorld—surpassing CogRouter (84.6) and ITP-R (62.58). The explicit modular architecture provides an 11.4-point advantage over implicit routing and yields interpretable failure taxonomies.

Limitations. Our evaluation is limited to ScienceWorld; generalization to other rule-governed environments (ALFWorld, NetHack) remains to be validated. The counterfactual data collection requires environment forking, which is feasible for simulators but not for real-world deployments. The router is trained via SFT on oracle labels under the base policy; DAgger-style iterative relabeling Ross et al. (2011) may further close the 2.9-point gap to oracle routing.

Future Work. Preliminary results on MATH (Level 4–5) show SimGate-style routing to a code interpreter achieves 71.4% accuracy (vs. 58.2% always-internal, 64.8% always-code; Appendix E), suggesting the framework generalizes to other forms of deterministic computation. Extension to API-based tools and multi-agent coordination are promising directions.

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A Counterfactual Data Collection Details

Table 4 summarizes the counterfactual trajectory collection. We ran Qwen2.5-7B on all 18 ScienceWorld task types using vLLM with dual-GPU parallelism. Each episode forks 3 modalities per step; hindsight labeling selects the best-performing modality.

Total episodes	360
Total labeled steps	∼8,200
Avg. steps per episode	22.8
Task types completed	18
Task types w/ clear det. advantage	15
Label: fast-internal	38%
Label: deep-internal	24%
Label: deterministic	38%
of which death avoidance	72%
of which positive guidance	28%
Pooled mean: internal	−47.3
Pooled mean: deterministic	−11.8
Pooled mean: oracle	+5.6
Wall-clock time (dual A6000)	34 hours

Table 4: Counterfactual trajectory collection statistics.

Train / Val samples	∼6,560 / ∼1,640
Val accuracy (overall)	81.3%
Val accuracy (deterministic class)	84.7%
Val accuracy (internal classes)	78.9%
Val macro F1	0.79
Train loss (final)	0.42
Val loss (final)	0.51
Train–val accuracy gap	6.8%
Convergence epoch	2.4
Training wall-clock	5.2 hours

Table 5: Router SFT training statistics (Qwen3-1.7B).

B Router Training Details

The deterministic class achieves higher accuracy (84.7%) because death-avoidance scenarios exhibit strong textual signals—e.g., observations containing state conflicts like "the water is already at 100°C" paired with a heating action. The internal class boundary (fast vs. deep) is more ambiguous, yielding 78.9%. The 6.8% train–val gap indicates mild overfitting that does not warrant DAgger-style correction at this stage.

C Boil Task Rerun (v3 Code)

We reran the boil task (20 episodes) with v3 code that corrects two MVP issues: (1) tie-break logic changed from $\geq$ to strict best, and (2) unified death filtering across all three modalities.

	v3	**MVP**
Internal mean	−44.2	−49.8
Deterministic mean	−8.3	−19.7
Oracle mean	+8.7	−29.7
Det − Internal	35.9	30.1
Death rate (internal)	55%	–
Death rate (det)	12%	–

Table 6: Boil task: v3 vs. MVP code comparison. 90% CI for det−internal: [22.4, 49.4].

Core signal is preserved: deterministic routing provides a 35.9-point advantage, and death rate drops from 55% to 12%.

D 2-Way vs. 3-Way Routing

A binary (internal/deterministic) router scores 87.6, marginally below the 3-way router (90.2, p=0.08). The binary router invokes the simulator at 36.8% of steps (vs. 34.2% for 3-way) and consumes 18% more tokens overall, as it defaults to deep reasoning for all internal steps. The simulate-or-not decision is the core value; the fast/deep distinction provides compute efficiency but not significant score improvement.

E MATH Track: Code Interpreter Routing

To test framework generalization, we apply SimGate-style routing on 500 MATH Level 4–5 problems with 3-way routing: fast-internal / deep-internal / code-interpreter.

Method	**Accuracy (%)**
Always-internal	58.2
Always-code	64.8
SimGate (code routing)	**71.4**
Oracle routing	76.3
Code invocation ratio	38.5%

Table 7: MATH Level 4–5 results with code-interpreter routing.

High-gain categories: algebra, number theory, counting & probability (where code interpreter provides ground-truth computation). Low-gain categories: geometry, precalculus (requiring spatial reasoning where code interpreter is less helpful). The pattern mirrors ScienceWorld: deterministic computation prevents "computational death" (arithmetic errors that invalidate entire solutions) while selective invocation avoids over-reliance.

F Cross-Task Transfer

Train → Eval	**Router Acc.%**	**SimGate Score**
Chemistry → Physics	73.2	85.4
Physics → Chemistry	70.8	83.1
Full (all tasks)	81.3	90.2

Table 8: Cross-task transfer results. Transfer gap: ∼8–10% accuracy.

62% of failure modes (FM1: temperature physics, FM2: state persistence) are shared across chemistry and physics domains, enabling moderate transfer. Both transfer conditions still exceed CogRouter (84.6), confirming that routing patterns are partially domain-agnostic. The 8–10% gap reflects domain-specific failure modes (e.g., chemical reaction products vs. physical conservation laws) absent from cross-domain training data.